Web Development Assignment Documentation

Detailed Report on HTML5 & CSS3 Front-End Web Modules

Subject: Web Development Lab / Theory	**Submitted For:** Academic Assignment Evaluation
Technologies Used: HTML5, Semantic Elements, CSS3, Flexbox | **Modules Documented:** 3 Web Projects (College, Travel, Food)	

1. Executive Summary & Assignment Scope

This technical documentation outlines three distinct front-end web applications developed as part of practical web development coursework. The objective of the assignment is to demonstrate strong proficiency in semantic HTML5 markup, modular CSS3 styling, layout architecture (CSS Flexbox and responsive box-model mechanics), accessible form design with client-side attributes, and interactive user-experience (UX) elements without reliance on external frameworks.

2. Module 1: NRI Institute of Technology – Institutional Portal

Source File Reference: `ass-2.html` | **Focus:** Informational Hierarchy, Educational Showcase & Responsive Cards

2.1 Module Overview & Objectives

Purpose: Provides a multi-section institutional landing page showcasing academic vision, departmental offerings, faculty breakdown, admissions criteria, and college contacts.
Target Audience: Prospective students, enrolled engineering candidates, guardians, and academic visitors.

2.2 Key Features & Semantic Layout

- **Sticky Header & Global Navigation:** Styled with brand gradient (#123c69 to #2563a6) featuring quick-jump anchor links to #home, #about, #courses, #faculty, #admissions, #contact, and external portfolio.
- **Hero Billboard:** High-impact visual banner highlighting core motto ('Education | Innovation | Excellence') with direct Call-To-Action (CTA) leading to admissions.
- **Card-Based Information Display:** Multi-column Flexbox grid rendering institutional highlights (Modern Learning, Computer Labs, Student Activities) with hover state elevation.
- **Curriculum Catalog:** Distinct card representations for 5 primary engineering streams: B.Tech CSE, ECE, EEE, Mechanical, and Civil Engineering.
- **Structured Admissions Flow:** Outlines statutory documentation requirements using clean unordered list typography.

2.3 Technical & CSS Implementation Highlights

Layout Paradigm: CSS Flexbox (`display: flex; flex-wrap: wrap; justify-content: center; gap: 22px;`) ensuring smooth responsiveness across varying viewport widths.
Smooth Navigation: Configured `html { scroll-behavior: smooth; }` for frictionless in-page anchor traversal.
Color Palette: Deep Navy Blue (#123c69), Vibrant Teal Accent (#087f8c), Warm Amber CTAs (#ffb703 / #fb8500), Soft Background (#f4f7fb).

3. Module 2: TravelGo – Bus and Train Ticketing Portal

Source File Reference: `ass-4.html` | **Focus:** Multi-Stage Booking Flow, Interactive Layouts & Tabular Pricing

3.1 Module Overview & Objectives

Purpose: Simulates an end-to-end transport booking portal enabling passengers to search departures, evaluate intercity transport options, input traveller verification data, inspect interactive seat maps, and review route fares.
Target Audience: Intercity commuters travelling key South Indian corridors (e.g., Vijayawada to Hyderabad).

3.2 Key Features & Architectural Modules

- **Booking Query Form:** Interactive query form capturing departure city, destination, journey date, travel mode (Bus/Train), and passenger headcount.
- **Comparative Service Cards:** Split-service display detailing bus schedule versus train schedule with departure/arrival indicators.
- **Passenger Verification Module:** Validated form fields including Full Name, Age constraints (min 1, max 120), Gender selection, Email, and 10-digit mobile number pattern validation (`pattern='[0-9]{10}'`).
- **Visual Bus Seat Layout (Pure HTML/CSS):** Simulated bus floorplan with driver cabin indicator, walking aisle separation, and color-coded statuses (Available vs. Booked) with matching key legend.
- **Fare Comparison Matrix:** Semantic HTML data table with alternating row zebra-striping and hover highlighting.

3.3 Fare Structure Matrix

Travel Type	Class / Tier	Sample Scheduled Fare
Bus	AC Sleeper	₹850
Bus	Non-AC Seater	₹500
Train	Chair Car (CC)	₹450
Train	Sleeper Class (SL)	₹300

4. Module 3: Campus Cravings – Quick-Service Food Delivery

Source File Reference: `food.html` | **Focus:** E-Commerce Catalog, Digital Ordering Flow & Campus Logistics

4.1 Module Overview & Objectives

Purpose: Serves as a campus food delivery storefront designed for college students and faculty desiring fast deliveries to hostel blocks or academic departments.

4.2 Key Features & Digital Ordering Workflow

- **Hero Billboard & Conversion CTA:** Engaging header with prompt direct anchor links driving users straight to the active food menu.
- **Student-Friendly Menu Catalog:** Structured card components featuring student-budget items with dish descriptions, bold price badges, and 'Order Now' action buttons.
- **Order Dispatch Form:** Comprehensive checkout form with fields for customer name, validated contact number, campus hostel/block drop location, menu selection dropdown, item quantity (1–10), and payment method radios.
- **Payment Flexibility:** Dual options supporting Cash on Delivery (COD) as default and instant Digital UPI / QR Code transfer.

4.3 Cafeteria Menu & Price Breakdown

Item Name	Description & Ingredients	Price (INR)
Crispy Veg Burger	Crunchy potato patty layered with fresh lettuce, mayo, and toasted buns	₹60
Paneer Tikka Roll	Char-grilled spiced paneer cubes rolled inside a warm wheat paratha	₹90
Margherita Pizza	Classic hand-stretched crust topped with fresh tomato sauce and melted mozzarella	₹149
Peri Peri French Fries	Crispy golden potato fries tossed with a zesty, spicy peri-peri seasoning	₹50
Cold Coffee	Thick, refreshing blended coffee served chilled with chocolate syrup drizzle	₹45
Schezwan Veg Fried Rice	Wok-tossed basmati rice with crunchy seasonal veggies in spicy schezwan sauce	₹80

5. Comparative Analysis of Web Engineering Concepts

Evaluation Metric	ass-2.html (College)	ass-4.html (TravelGo)	food.html (Campus Cravings)
Primary Use Case	Informational & Institutional	Ticketing & Reservation System	On-Campus E-Commerce Ordering
Layout Mechanism	CSS Flexbox (Wrapping grid)	Flexbox (Cards, Seats, Nav)	CSS Grid / Flexbox Cards
Form Handling	Basic Contact Redirection	Search & Passenger Details Form	Detailed Order Form with Payment
Interactive UI Elements	Hover Card Elevation, Anchor Nav	Visual Seat Map, Fare Matrix	Menu Price Badges, Dropdown Select

6. Key Takeaways & Academic Reflections

Semantic Accessibility: Proper utilization of HTML5 structural elements (`<header>`, `<nav>`, `<section>`, `<article>`, `<main>`, `<footer>`) ensures high screen-reader compatibility and SEO best practices.

Form Usability & Client-Side Verification: Employing strict HTML5 attributes (required, pattern, min, max, type='email', type='tel') delivers real-time validation feedback.